Course Project

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Install packages

```
library(dplyr)

library(xts)

library(PerformanceAnalytics)

library(zoo)
```

Data Cleaning

Load data

```
data <- read.csv("~/Desktop/work stuff/Imperial/R/cw/compustat_daily_2010_2025.csv")
```

Select PFE from data

```
PFE_data <- data %>%
  mutate(datadate = as.Date(datadate, format = "%m/%d/%Y")) %>%
  filter(tic == "PFE") %>%
  select(tic, datadate, conm, prccd) %>%
  arrange(datadate)

head(PFE_data)
```

```
##   tic   datadate       conm prccd
## 1 PFE 2010-01-04 PFIZER INC 18.93
## 2 PFE 2010-01-05 PFIZER INC 18.66
## 3 PFE 2010-01-06 PFIZER INC 18.60
## 4 PFE 2010-01-07 PFIZER INC 18.53
## 5 PFE 2010-01-08 PFIZER INC 18.68
## 6 PFE 2010-01-11 PFIZER INC 18.83
```

Autocorrelation Functions

ACF of Daily Returns

```
PFE_data <- PFE_data %>%
  arrange(datadate) %>%
  mutate(daily_returns = prccd/lag(prccd) - 1) %>%
  na.omit()

daily_acf <- acf(PFE_data$daily_returns, main = "Autocorrelation of PFE Daily Returns")
```

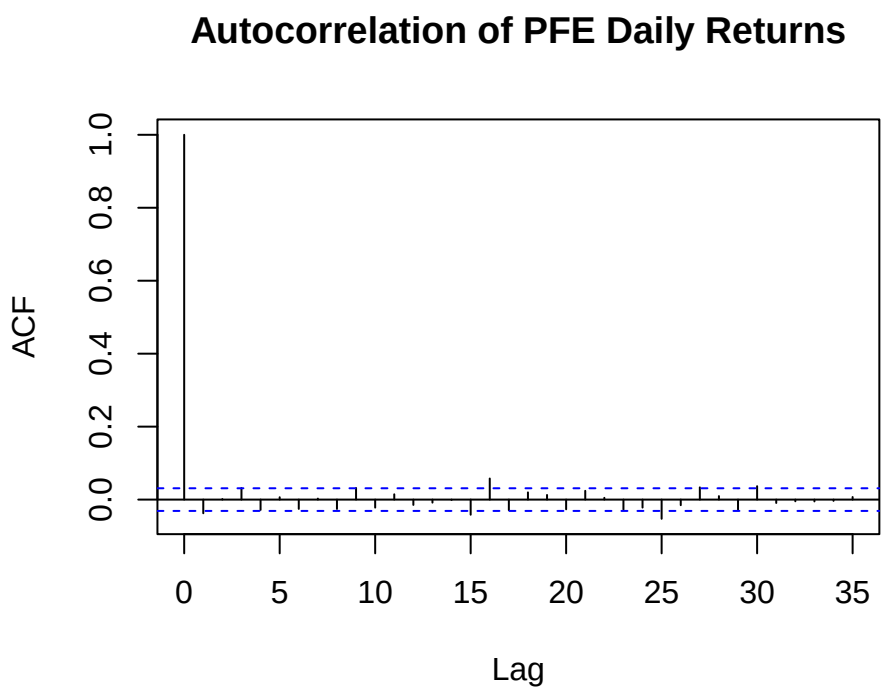

Figure 1: Autocorrelation of PFE Daily Returns

ACF of Average Returns over Weekly Mean Returns

```
PFE_weekly <- PFE_data %>%
  mutate(week = format(datadate, "%Y-%U")) %>%
  group_by(week) %>%
  summarise(weekly_returns = mean(daily_returns), na.rm = TRUE) %>%
  na.omit()

weekly_acf <- acf(PFE_weekly$weekly_returns, main = "Autocorrelation of PFE Weekly
↪  Returns")
```

Identify Statistically Significant Autocorrelation

```
cat("Significant daily autocorrelations (> 0.2):\n")
```

```
## Significant daily autocorrelations (> 0.2):
```

```
print(which(abs(daily_acf$acf) > 0.2))
```

```
## [1] 1
```

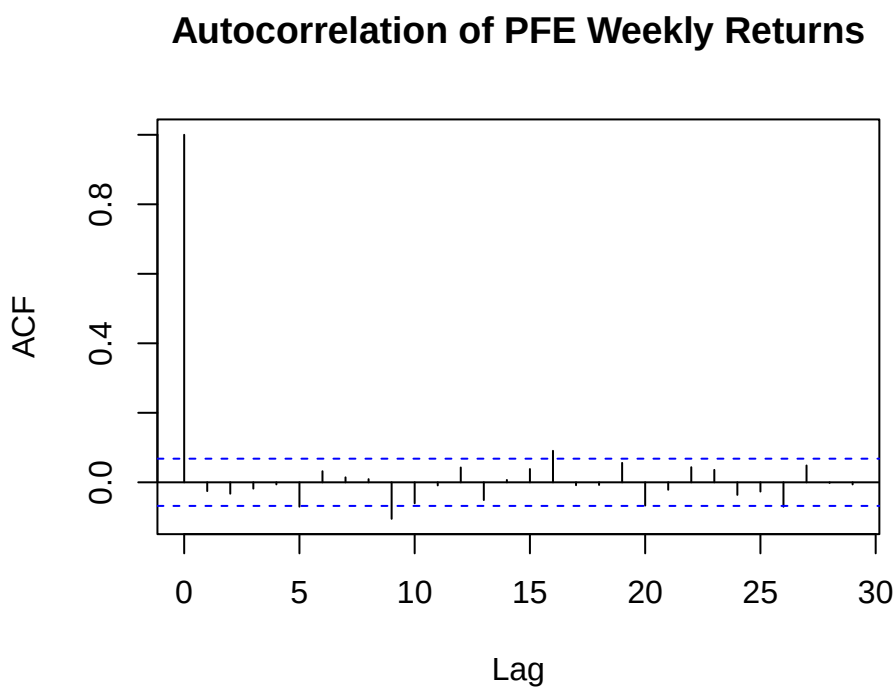

Figure 2: Autocorrelation of PFE Weekly Returns

```
cat("Significant weekly autocorrelations (> 0.2):\n")
```

```
## Significant weekly autocorrelations (> 0.2):
```

```
print(which(abs(weekly_acf$acf) > 0.2))
```

```
## [1] 1
```

Moving Average Crossover Strategy

Create a Subset of the Data

```
strategy_subset <- PFE_data %>%
  arrange(datadate) %>%
  filter(datadate >= as.Date("2020-01-01")) %>%
  na.omit()
```

Generate Buy and Sell Signals

```r
source("functions.r")

strategy_subset <- strategy_subset %>%
  mutate(fast_ma = moving_avg(prccd)) %>%
  mutate(slow_ma = moving_avg(prccd, 200)) %>%
  mutate(row_num = row_number())

PFE_xts <- xts(strategy_subset$prccd, strategy_subset$datadate)

fast_MA <- rollapplyr(PFE_xts, width = 50, FUN = mean)
slow_MA <- rollapplyr(PFE_xts, width = 200, FUN = mean)
signals <- sign(fast_MA - slow_MA)
signals[is.na(signals)] <- 0
names(signals) <- "signal"
head(signals)
```

```
##            signal
## 2020-01-02      0
## 2020-01-03      0
## 2020-01-06      0
## 2020-01-07      0
## 2020-01-08      0
## 2020-01-09      0
```

Plot Strategy Performance

```r
charts.PerformanceSummary(backtest(PFE_xts, signals),
                          main = "Market vs Strategy")
```

Figure 3: Strategy Performance

Compute Numeric Performance Strategy

```
market_vs_strategy <- backtest(PFE_xts, signals)

cumulative_market <- tail(cumprod(1 + market_vs_strategy$Benchmark) - 1, 1)
cumulative_strategy <- tail(cumprod(1 + market_vs_strategy$Strategy) - 1, 1)

paste("Market return is", round(cumulative_market * 100, 2), "%")
```

```
## [1] "Market return is -37.94 %"
```

```
paste("Strategy return is", round(cumulative_strategy * 100, 2), "%")
```

```
## [1] "Strategy return is -8.27 %"
```

Bollinger Bands Breakout Strategy

AI Generated Strategy

```
# AI-Generated Bollinger Bands Breakout Strategy Function
```

```r
bollinger_bands_breakout <- function(prices, n = 20, k = 2) {
  # Calculate Bollinger Bands
  # Middle Band = n-period simple moving average (SMA)
  # Upper Band = SMA + k * standard deviation
  # Lower Band = SMA - k * standard deviation

  # Input validation
  if(length(prices) < n) {
    stop("Insufficient data points for the specified window size")
  }

  # Initialize signals vector
  signals <- rep(NA, length(prices))

  # Calculate Bollinger Bands
  for(i in n:length(prices)) {
    # Get the current window
    window_prices <- prices[(i-n+1):i]

    # Calculate middle band (SMA)
    middle_band <- mean(window_prices)

    # Calculate standard deviation
    sd_band <- sd(window_prices)

    # Calculate upper and lower bands
    upper_band <- middle_band + k * sd_band
    lower_band <- middle_band - k * sd_band

    # Current price
    current_price <- prices[i]

    # Generate signals
    if(current_price > upper_band) {
      # Buy signal when price breaks above upper band
      signals[i] <- 1
    } else if(current_price < lower_band) {
      # Sell signal when price breaks below lower band
      signals[i] <- -1
    } else {
      # Hold signal when price is within bands
      signals[i] <- 0
    }
  }

  return(signals)
}
```

Apply Bollinger Bands strategy to PFE Close Prices

```r
# Apply Bollinger Bands strategy to PFE data
bb_signals <- bollinger_bands_breakout(as.numeric(PFE_xts), n = 20, k = 2)
```

Perform Backtesting

```
# Convert signals to xts format for backtesting
bb_signals_xts <- xts(bb_signals, index(PFE_xts))

# Backtest Bollinger Bands strategy
bb_performance <- backtest(PFE_xts, bb_signals_xts)
```

Plot performance summary

```
charts.PerformanceSummary(bb_performance, main = "Bollinger Bands Breakout Strategy
↪  Performance")
```

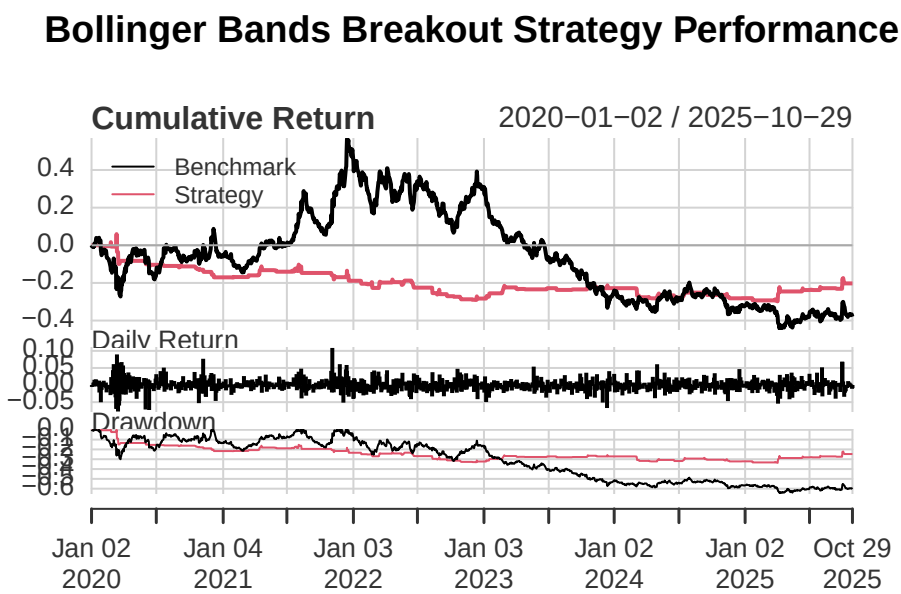

Figure 4: Bollinger Bands Breakout Strategy Performance

Analysis

The autocorrelation analysis revealed minimal persistence in Pfizer's daily and weekly returns, with no statistically significant autocorrelations exceeding the 0.2 threshold in either time frame. Only lag 1 showed significance due to the nature of ACF calculation. This suggests that PFE's price movements exhibit weak-form market efficiency, where past price information provides little predictive power for future returns.

The AI-generated Bollinger Bands breakout strategy employed a conventional approach using 20-period moving averages with 2 standard deviation bands. The strategy generated buy signals when prices breached the upper band (interpreted as momentum continuation) and sell signals when crossing below the lower band. This approach aligns with standard technical analysis literature, though some sources suggest alternative interpretations of Bollinger Band breakouts. The performance was similarly suboptimal, likely due to the strategy's sensitivity to volatility spikes and frequent whipsaws in the turbulent post-2020 market environment.

Both strategies demonstrated the limitations of technical analysis in efficient markets, particularly during periods of high volatility. The findings suggest that while mechanical trading rules can capture some market inefficiencies, their performance is heavily dependent on market conditions and may struggle during transitional or highly volatile periods.